

Investigation of optical properties and chemical structure of Nd₂O₃ nanoparticles deposited on NaX zeolite powder using plasma focus device

Mir Mohammad Reza Seyedhabashi (✉ m.seyedhabashy@gmail.com)

Nuclear Science and Technology Research Institute (NSTRI)

Hesham Hosseinpour

University of Isfahan

Ehsanollah Noori

Nuclear Science and Technology Research Institute (NSTRI)

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Abstract

In this study, nanoparticles of neodymium oxide were deposited on NaX zeolite powder using a 5.3 kJ Mather–type plasma focus device. The NaX zeolite samples were exposed to the 10, 15, and 20 shot numbers of the plasma focus device at a distance of 14 cm from the anode tip and 0 angular positions with respect to the anode axis. Parent NaX zeolite and NaX:Nd composites with different Nd_2O_3 percent weights were characterized using different techniques such as; XRF, XRD, SEM, FT–IR, and UV–vis. It was shown that the amount of Nd_2O_3 in the samples increases with increase of the shot number. It was also found from the results that variations in amount of Nd_2O_3 can have significant effects on the degree of the crystallinity and optical properties of final products such as refractive index, band gap energy, dielectric constant, and absorption coefficients.

1. Introduction

Neodymium is an exclusive element that can be employed as a luminescent center in the laser systems because of the four-level energy structure which is desirable for laser transitions[1]. Extensive studies have been conducted with the applications of neodymium in photonic devices [2, 3]. The $4f^N$ electronic level structures of these ions provide many long-lived intermediate levels which can be populated using infrared radiation. In addition, there are some metastable higher lying levels resulting in visible strong emissions. The rare-earth oxide Nd_2O_3 has been attracted much attention because of its high band gap (5.8 eV), high dielectric constant (10–15), favorable lattice matching, and proper thermodynamic stability in conditions of contact with silicon substrates. For these reasons, several studies have been focused on neodymium doped materials and their possible applications in related telecommunication and laser facilities such as optical amplifiers and solid state lasers [4–8].

So far, various methods such as chemical approaches, physical thin film techniques, pulsed laser deposition (PLD), radio frequency (RF) sputtering, and molecular beam epitaxy (MBE) have been proposed to deposit nanoparticles on different substrate materials [9]. Recently, because of several advantages in comparison with the mentioned techniques, application of plasma focus (PF) device in deposition of nanoparticle has been developed greatly. The merits of using plasma focus device over the other methods include low gas consumption, elimination of the need for additional heating of the specimen, very short time compared with DC pulse or RF glow discharge, and convenience and ease of device maintenance under pressure conditions [10].

Zeolites are crystalline aluminosilicates constructed from SiO_4 and AlO_4 tetrahedra which are joined to each other by sharing all of the oxygen atoms. Zeolites have wide range of applications, such as catalysis, ion exchange, and separation [11, 12]. In recent years, possible applications of zeolites as host substances for luminescent guest species has been investigated and corresponding results have verified zeolites and related structures as promising new classification of luminophores [13]. Although PF device has been widely used for the deposition of nanoparticles on various substrates, thorough and detailed studies of nanoparticles deposition on powdery substrates has not been carried out yet. The main concern of the present study is to investigate the effect of PF device shot numbers on the Nd_2O_3 nanoparticle deposited on NaX zeolite powder. To do this, the effect of the shot number on the optical properties and chemical structure of the prepared NaX/ Nd_2O_3 composite was studied in detail. The organization of the paper is as follows: In Sec. 2 experimental methodology, including introduction of the materials which are used in the experiments, zeolite synthesis, preparation of the Nd_2O_3 composite, and characterization of the samples are explained. Sec. 3 is devoted to the results and discussion of the XRD, XRF, FT-IR and SEM analysis of the samples, and conclusion is made in Sec. 4.

2. Experimental Methodology

2.1. Materials

All materials were utilized as received with no any further purification. The silicon source that utilized in zeolite synthesis was sodium silicate ($\text{Na}_2\text{SiO}_3 - 9\text{H}_2\text{O} > 98\%$, Sigma – Aldrich), and the aluminum source was sodium aluminate (54.3% Al_2O_3 , 44.5% Na_2O , Riedel–de – Haen), pellets of sodium hydroxide were also used as the sodium source (99%, Sigma – Aldrich).

2.2. NaX zeolite Synthesis

In summary, synthesis process of NaX zeolite is as follows: first, 4.2 gr of $\text{Al}(\text{OH})_3$ is dissolved in 8.4 mL of NaOH aqueous solution (50% w/v), then 8.5 mL of 100 °C water is added. Later on, 15.1 gr of NaOH is poured into the mixture, once more, 60 mL water is added and is heated at 50 °C and therefore, the Solution No. 1 is obtained. In the next step, 45.2 gr of Na_2SiO_3 is dissolved in the mixture of 250 mL water and 15.2 gr NaOH to obtain the solution No. 2. In the next step, the solutions 1&2 are mixed quickly and the resulting solution is heated at 70 °C by an oven. At the last step, the resulting white powder which is used as parent NaX sample, is washed with water and dried at 80 °C [14].

2.3. Preparation of NaX: Nd₂O₃ composite

In order to investigate the optical properties of Nd nanoparticles deposited on the zeolite powder, 10, 15 and 20 focused shots of a Mather-type plasma focus (MTPF) device are used. The detailed description about construction and operation of PF device can be found extensively in literature [15–20]. The MTPF device of present study, as is shown schematically in Fig. 1, has 5.3 kJ energy and is powered by a 12μF, 30kV fast discharging energy storage capacitor.

A sharp spike in the current signal which is displayed on the screen of a digital storage oscilloscope is an indication of appropriate focusing condition. The anode of the MTPF device is designed as a hollow removable cylinder and hence, the neodymium plate of 99.99% purity can be fitted to the anode. The chamber of plasma is evacuated and filled with air to a pressure of 1.2 mbar. The substrate, on which the nanoparticle is to be deposited, is placed under electrode assemblies and is inserted through the bottom of the focus chamber.

2.4. Characterization

To characterize the samples, different techniques including: X-ray diffraction (XRD) (Bruker, D8ADVANCE) with Co–ka radiation, X-ray fluorescence (XRF) (Bruker, S4 PIONEER), and scanning electron microscopy (SEM) (TESCAN, VEGA II) are considered. The infrared transmission spectrum (FT–IR) is also recorded by an FTIR spectrometer system 2000 FT–IR (Perkin–Elmer) using the KBr wafer method.

3. Results and Discussion

3.1. XRD analysis

The XRD patterns for the synthetic NaX zeolite powder and NaX:Nd composite with various Nd₂O₃ nanoparticle concentrations are shown in Fig. 2.

In the case of NaX zeolite, the characteristic peaks at the 2θ values of 6.93, 11.44, 13.45, 17.79, 21.28, 23.22, 27.01, 30.94, 35.23, 36.01, 37.25 and 39.15° correspond to the Miller indices (h k l) of (1 1 1), (2 2 0), (3 1 1), (3 3 1), (5 1 1), (4 4 0), (5 3 3), (6 4 2), (6 6 0), (5 5 5), and (6 6 4), respectively. The aforementioned results are in good agreement with those of the face-centered cubic crystal structures of NaX zeolite (molecular formula: C₅H₄O₂ .Na₂O. Al₂O₃ 3.3SiO 2.7H₂O) (Pdf No.41–118) with the lattice parameter of a = 24.96000 (Å) and 1.92 (g/cm³) measured density. It is clear from the figure that the zeolite structure is remained well maintained after the shots. By following a standard procedure which is described in Ref. [21], the zeolite crystallinity is estimated from the ratio of the sums of the intensities of the peaks of (1 1 1), (2 2 0), (3 1 1), (3 3 1), (5 1 1), (4 4 0), (5 3 3), (6 4 2), (6 6 0), (5 5 5), and (6 6 4) pertaining to the NaX:Nd composite to those of the reference material (pure zeolites). A slight decrease in crystallinity is found as the shot number of the PF device increases. The founding results are sammarized in Table 1 and can be compared with the pure zeolite crystallinity given in Fig. 2.

Table 1 The values of the crystallinity for the samples		
Sample	Peaks Intensity	Crystallinity (%)
NaX	324.45	100
NaX:Nd1	263.51	81.2
NaX:Nd4	236.79	72.9
NaX:Nd5	233.75	72.0

3.2. XRF studies

Table 2 describes the constituent elements of the NaX zeolite composition which are determined by XRF analysis of the samples. The essential parameter that determines the capability of zeolites as an efficient absorbent material is Si/Al ratio which is approximately 1.78 for prepared NaX zeolite.

Table 2
Elemental analysis of NaX zeolite powder

Compound	Si	Al	Na	Cl	Ca	S	Fe	Cu	Ba	K	Sr	Zn	*LOI	Total
Concentration (%W/W)	41.55	23.40	11.60	6.10	5.38	0.375	0.159	0.117	0.110	0.092	0.65	0.32	12.42	101.4
*Loss on Ignition (1000°C, 2 h)														

The original oxides are determined by XRF analysis after 10, 15 and 20 shots of the MTPF device and are summarized in Table 3.

Table 3
The values of XRF analysis of NaX:Nd composite powder

Samples				
Compound	NaX	NaX:Nd1	NaX:Nd4	NaX:Nd5
SiO ₂	45.3	44.5	44.1	43.6
Al ₂ O ₃	34.4	34.7	33.9	33.1
Na ₂ O	15.9	16.6	15.7	15.5
Nd ₂ O ₃	0.06	0.7	1.6	3.1
Concentration (Wt. %)				

As it is understood from the XRF analysis, by increasing the shot number of the MTPF device, the amount of Nd₂O₃ also increases from 0.06–3.1%. The effect of increasing shot numbers is shown in Fig. 3

The concentration of the rare–earth ions provides an insight into the atomic arrangements, and it is one of the most important parameters as it affects the laser gain of the host material and the physical properties. The physical parameters such as the concentration of neodymium ions (N) can be calculated from the average molecular weight, density, and the concentration of the rare–earth ions and are expressed in the following form [22].

$$N(\text{ion}/\text{cm}^3) = \frac{(\% \text{mole of Nd}_2\text{O}_3) \times (\text{Avogadro's number}) \times (\text{zeolite density})}{(\text{zeolite average molecular weight})} \quad (1)$$

The calculated concentrations of neodymium ions are given in Table 4.

Table 4
The values of neodymium ion concentration for given samples

Sample	NaX	NaX:Nd1	NaX:Nd4	NaX:Nd5
Wt. (%)	0.06	0.7	1.6	3.1
Mole (%)	0.013	0.157	0.360	0.695
N(ion/cm ³)×10 ²⁰	1.655	4.328	9.92	19.159

3.3 FT-IR studies

Nd₂O₃ particle, NaX pure zeolite, and NaX:Nd composite considering various Nd₂O₃ nanoparticle concentrations are characterized by FT–IR. The corresponding results of the FT-IR analysis are shown in Fig. 4.

The FTIR spectrum of the synthesized Nd₂O₃ confirms that the fundamental mode of fluctuations below 500 cm^{–1} is due to the Nd–O stretching modes. The band at 1148 cm^{–1} indicates the saccharide structure at 851.9. The relevant peaks at 1364 and 1471 cm^{–1} are associated with the asymmetric stretching of the carboxyl group (C = O). The significant absorption peak of the samples at 3436 cm^{–1} is mainly because of O–H stretching [23–26].

Infrared spectra of absorption peaks of NaX zeolite at the range of $400\text{--}1200\text{ cm}^{-1}$, are related to the Si–O–Al, Si–O–Si, Al–O and Si–O–Na bands. The absorption peak at 462 cm^{-1} can be ascribed to the internal fluctuations of $(\text{Si, Al})\text{O}_4$ tetrahedral of zeolite X. The peaks at 569 and 755 cm^{-1} are attributed to the fluctuations of external bonds between tetrahedral, and sensitivity of the framework structure. The band which is concerned with 671 cm^{-1} , is because of symmetric stretching of the internal fluctuations of the $(\text{Si, Al})\text{--O}$. Fluctuations around 978 cm^{-1} can be explained in terms of a T–O stretch involving essential movements which are related to the oxygen atoms. Generally speaking, the bands in the wave number range $2500\text{--}3800\text{ cm}^{-1}$ and near 1650 cm^{-1} can be assigned to the different types of hydroxyl groups [27, 28].

As can be seen from the FT–IR analysis of the samples (NaX:Nd composite), no characteristic bands of crystalline Nd_2O_3 phase are found in the spectra which highlights similar results of NaX zeolite powder. This is due to low Nd_2O_3 content in NaX:Nd composite that causes a lower Nd_2O_3 fluctuation intensity in the NaX. These findings show that adding Nd_2O_3 does not has significant effect on the structure of NaX zeolite [28]. On the other hands, increasing the amount of Nd_2O_3 content, leads to the intensity reduction of some characteristic bands.

3.4. SEM images analyses

Microstructure evolution of the sample is presented in SEM micrographs. One may notice that Nd_2O_3 is not uniformly distributed throughout the zeolite matrix, leading to the repercussion by which the structure of the clusters becomes irregular. Pure NaX zeolite without any cluster in pores is shown in Fig. 5.

After 10, 15 and 20 shots, Nd_2O_3 is distributed in the pores and on the surface of zeolite as illustrated in Fig. 5 (b-d) respectively.

3.5. Optical absorption

The absorption spectra of samples with respect to the concentration of Nd_2O_3 are shown in Fig. 6.

It can be understood from the figure that increasing the concentration of Nd_2O_3 , leads to increase of absorption intensity, considering different shot numbers of the plasma focus device. On the other words, absorption coefficients which are associated with the absorption bands, are dependent on the concentration of the neodymium oxide. The transitions of: $^4\text{I}_{9/2} \rightarrow ^4\text{F}_{5/2} + ^2\text{H}_{9/2}$, $^4\text{F}_{7/2} + ^4\text{S}_{3/2}$, $^2\text{G}_{7/2} + ^4\text{G}_{5/2}$, $^2\text{K}_{13/2} + ^4\text{G}_{7/2} + ^4\text{G}_{9/2}$, $^4\text{D}_{3/2} + ^4\text{I}_{11/2} + ^4\text{D}_{5/2}$, $^4\text{D}_{1/2} + ^2\text{L}_{15/2}$ between the higher energy and ground states, inside the 4f^3 electronic configuration of the Nd^{3+} ions, are observed in the $400\text{--}900\text{ nm}$ region of the spectrum. It is also deduced from the figure that the associated wavelength of each absorption peak is not shifted with respect to the concentration of Nd^{3+} ions.

Generally speaking, refractive index plays very crucial role in analysis and in-depth study of lasing process of optical materials

Therefore, determination of refractive index, as an inter-disciplinary parameter, is essentially important in creating correlation between the theory of electronic structure and amorphous properties of materials. The refractive indexes of the samples can be calculated from the reflection spectra at transition wavelength in the following form [29]

$$n_d = \frac{1 + \sqrt{R}}{1 - \sqrt{R}}$$

2

Finding the refractive index, the dielectric constant of the samples can be expressed as [30]

$$\varepsilon = n_d^2$$

3

Increasing the refractive index leads to increase of the density of materials as well as dielectric constant. The calculated values of the reflection index and dielectric constant of samples are given in Table 5.

Table 5
The values of neodymium ion concentration and the values of optical parameters for samples

Sample	NaX	NaX:Nd1	NaX:Nd4	NaX:Nd5
N (ion/cm ³)×10 ²⁰	1.655	4.328	9.92	19.159
Mean absorption wavelength (λ_p) (for the $^4I_{9/2} \rightarrow ^4F_{5/2}$ transition)	803	803	803	803
Absorption cross-section (for $^4I_{9/2} \rightarrow ^4F_{5/2}$)	2.23015	0.83735	0.43752	0.27035
Refractive index (at transition wavelength: $^4I_{9/2} \rightarrow ^4F_{5/2}$)	1.26673	1.27237	1.28371	1.29768
Dielectric constant	1.6299	1.6187	1.6478	1.6837

The main reason of increase of the refractive index is strong packing of the rare earth materials. Allowed and forbidden transitions for direct and indirect optical band gaps can be calculated using Davis and Mott as [31, 32]

$$(\alpha h\nu)^n = B(h\nu - E_g)$$

4

where $h\nu$ is energy (frequency) of photon and E_g stands for optical energy of the band gap and B is introduced as a constant parameter. The exponent n equals to $1/2$ & 2 for the forbidden and allowed directed transitions, respectively. The optical energy of the band gap (E_g) for indirect and direct transitions is found by plotting $(\alpha h\nu)^{1/2}$ and $(\alpha h\nu)^2$ as functions of photon energy ($h\nu$) (Taucs plot). The plots of Taucs for the samples are shown in Fig. 7.

The respective values of E_g can be obtained by extrapolating the straight portion of the plot of $(\alpha h\nu)^{1/2}$ and $(\alpha h\nu)^2$ with respect to $h\nu$ [33, 34]. The obtained values for the optical energy of the band gaps are summarized in Tables 6 and 7.

Table 6
The values of direct forbidden transition band gap (eV), $n = 1/2$

Sample	Slope	Intercept	Band gap (eV)
NaX	1.85002	-1.02523	0.55417
NaX:Nd1	1.85003	-1.02527	0.55419
NaX:Nd4	1.8648	-0.95993	0.51476
NaX:Nd5	1.85332	-0.49823	0.26883

Table 7
The values of direct allowed transition band gap (eV), $n = 2$

Sample	Slope	Intercept	Band gap (eV)
NaX	2475.26129	-8264.07586	3.33866
NaX:Nd1	2475.28245	-8264.16707	3.33867
NaX:Nd4	2643.78994	-8805.27552	3.33055
NaX:Nd5	3097.26122	-10117.16635	3.26648

Evidently, the optical energy of the band gaps decreases with the increase in the concentration of neodymium ions. Such decrease in the energy of the band gap can be related to the progressive increase in the concentration of NBOs [35].

4. Conclusion

Neodymium and nanoclusters deposited on NaX zeolite powder were prepared by a novel sputtering method employing a Mather-type plasma focus device. The substrate was situated in 0 angular positions with respect to the anode axis, and was exposed to 10, 15 and 20 shots of the plasma focus device. Effects of shot number of the plasma focus device on the optical characteristics and structural properties of the deposited nanoclusters were investigated using XRD, XRF and SEM analysis, as well as FT–IR and absorption spectra of the samples.

The XRD results revealed that the crystallinity of the sample constantly decreases with the increase of the shot number from 10 to 20. The XRF analysis was considered to study the effect of shot number on the enhancement of the Nd_2O_3 values.

It was found from the FT–IR spectra of NaX:Nd composite samples that the content of Nd_2O_3 at NaX:Nd composite was low which consequently, caused a lower intensity fluctuation of the Nd_2O_3 in comparison with the NaX. According to the results, the content of Nd_2O_3 does not have any significant effect on the structural properties of the NaX zeolite. It was also found that the intensity of some characteristic peaks decreased with increasing the amount of Nd_2O_3 content. Microstructure evolution of the sample was observed in SEM analysis and showed irregular distribution in the pores and on the surface of the zeolite. Moreover, the increases of the refractive index and decrease in the energy of the band gap, revealed inversely proportionality to the concentration of neodymium ions. Our investigation opens up a new route in the area of dependency of the different properties of nanoclusters on the shot number of the plasma focus device. In all experiments, the zeolite powder was placed at a constant distance from the tip of the anode (140 mm). If the current work is to be extended and studied further, the authors will certainly confirm the impact of the distance of the film from the tip of the anode.

Declarations

Data availability statement:

Data will be made available on reasonable request

Author contribution statement:

All authors contributed to the study conception and design. Material preparation, data collection and analysis were performed by Mir Mohammad Reza Seyedhabashi, Hesham Hosseinpour, and Ehsanollah Noori. The first draft of the manuscript was written by Mir Mohammad Reza Seyedhabashi and all authors commented on previous versions of the manuscript. All authors read and approved the final manuscript.

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Figures

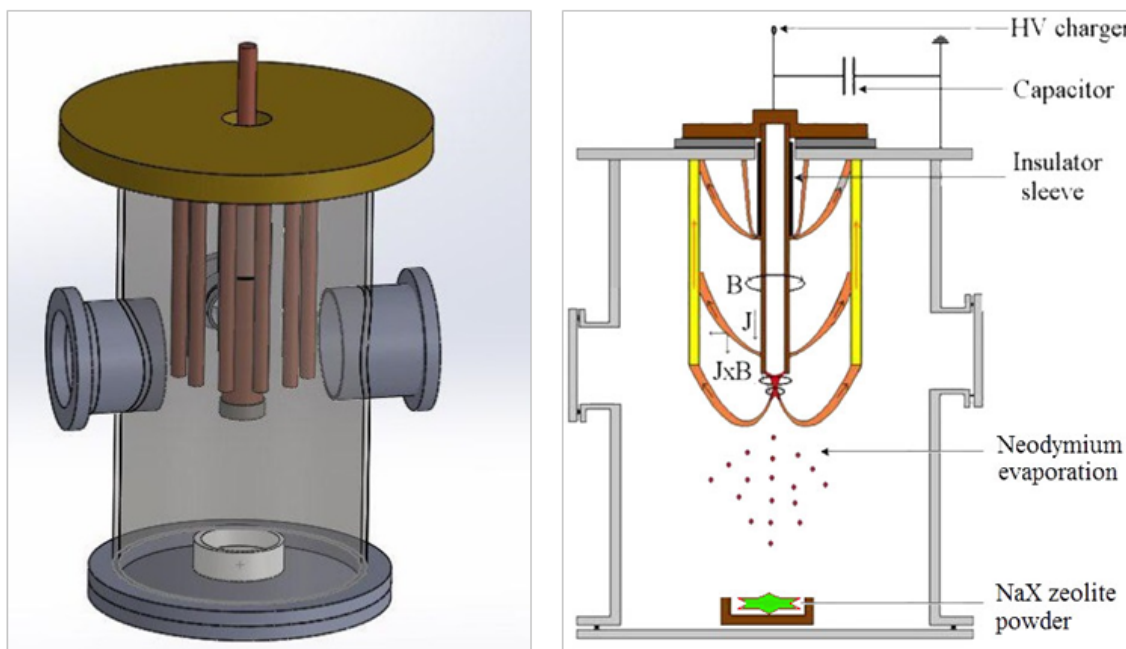


Figure 1

The schematic of the PF device and setup of the experiment

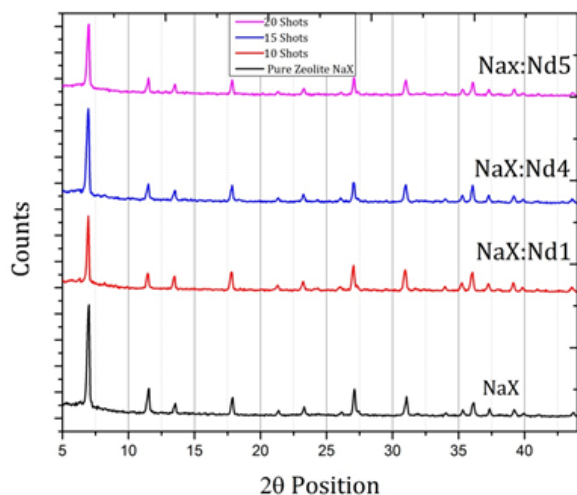


Figure 2

XRD pattern of synthesized NaX:Nd composite

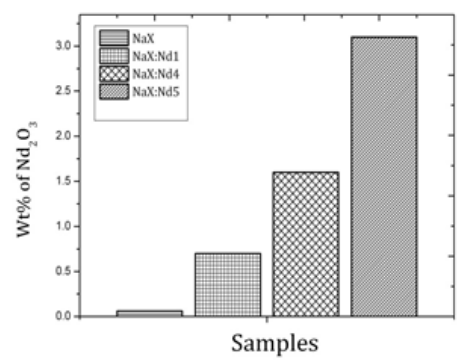


Figure 3

The effect of PF shot numbers on the Nd_2O_3 value

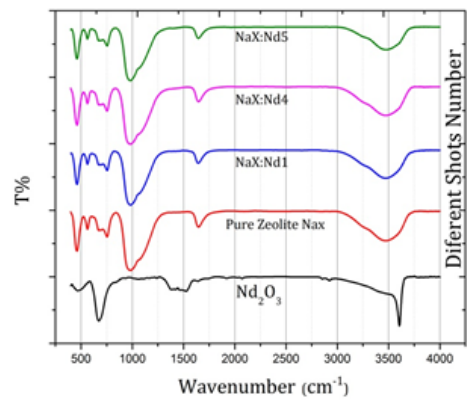


Figure 4

FT-IR spectra of ZnO nanoparticle, NaX Nanozeolite, and ZnO/NaX nanocomposite with various ZnO nanoparticle concentrations.

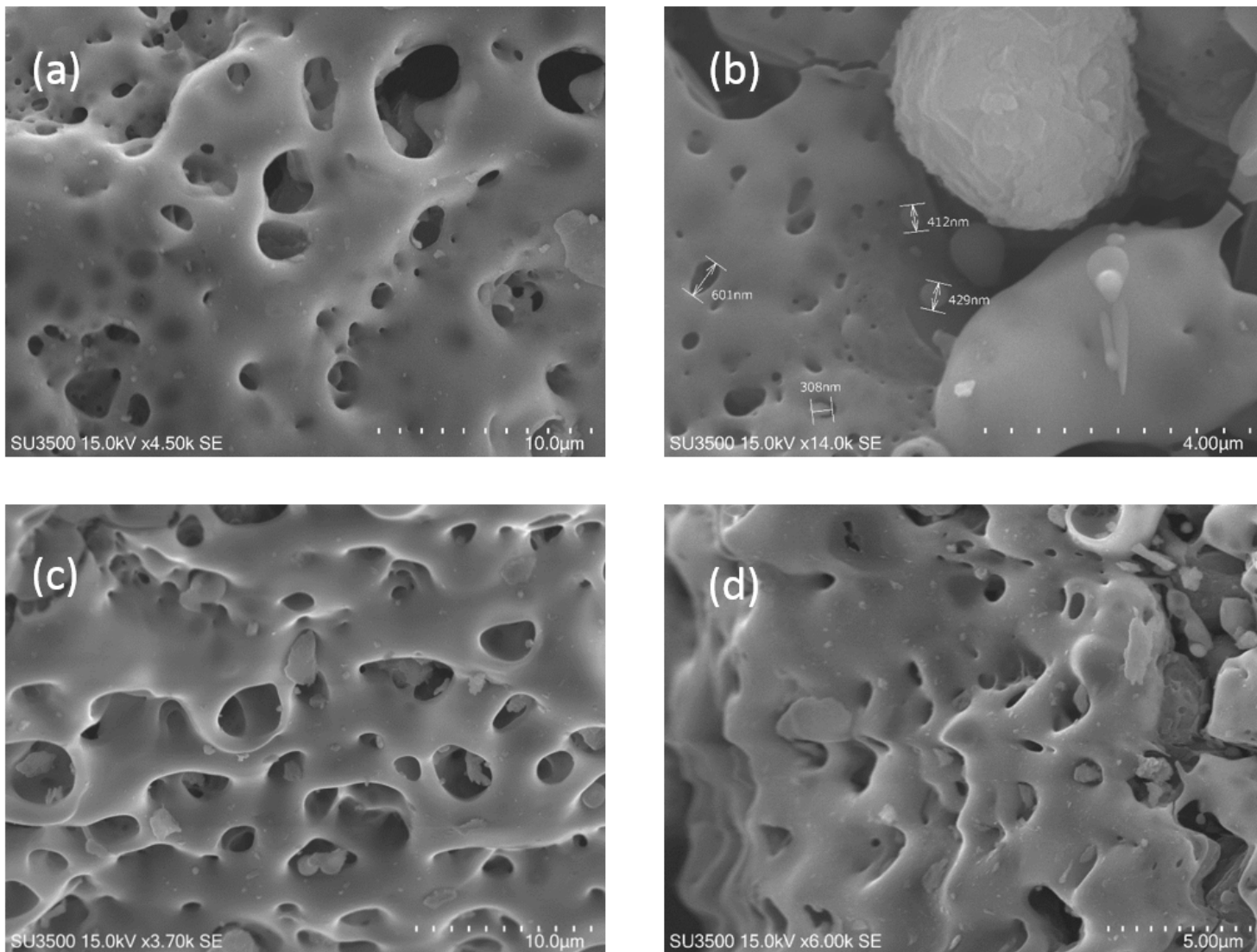


Figure 5
SEM micrograph of NaX Pure Zeolite (a), NaX:Nd1 (b), NaX:Nd4 (c), and NaX:Nd5

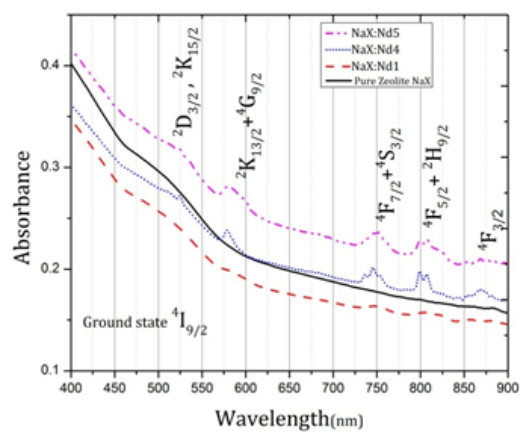


Figure 6
UV-vis absorption spectra of synthesized NaX:Nd composite

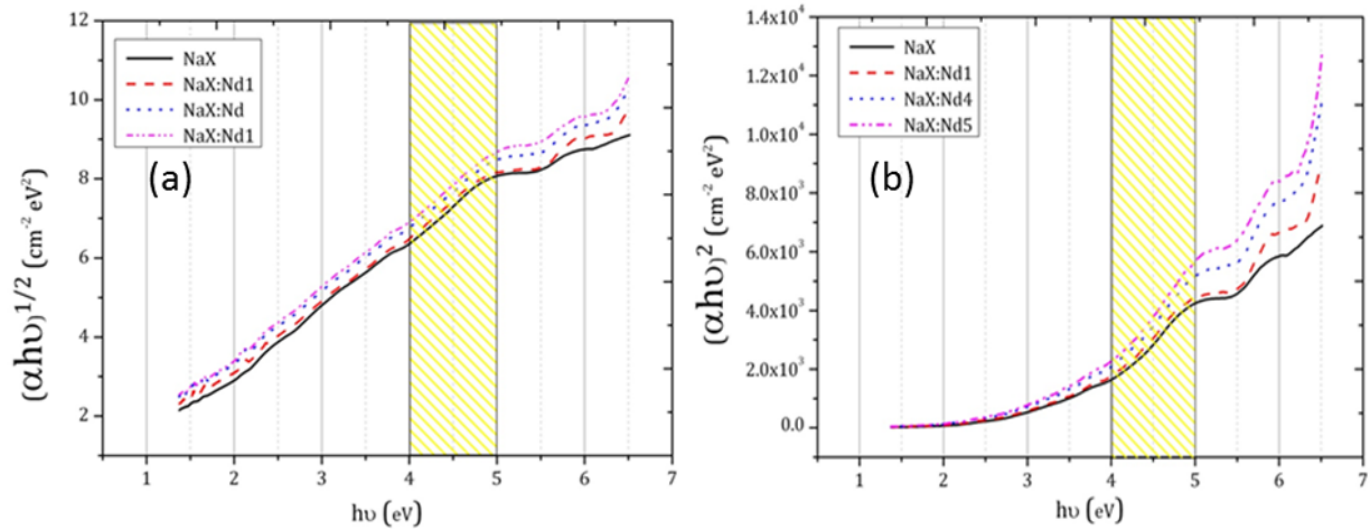


Fig. 7: Plots of $(\alpha h\nu)^{1/2}$ (a) and $(\alpha h\nu)^2$ as function of photon energy for all samples

Figure 7

See image above for figure legend.